MATHS MANTHAN

• TUITION CLASSES •

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Subject: Mathematics | **PERIODIC COMPREHENSIVE TEST – 09 (SOLUTIONS)** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 6: Triangles** | **MM: 30**

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit theorem/formula declaration, neat geometric diagrams, logical reasoning, and final boxed results.
2. In descriptive proofs, state **Given**, **To Prove**, **Construction** (where needed), and justify every equality or similarity criterion.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 1.5M, 2M]**. Alternative valid methods receive full credit.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Objective Questions

MATHS MANTHAN | (6 × 1 = 6 Marks)

Q1. If $\triangle ABC \sim \triangle DEF$ and AL, DM are their respective medians, then: [1]

(a) $\frac{AB}{DE} = \frac{AL^2}{DM^2}$ (b) $\frac{AB}{DE} = \frac{AL}{DM}$ (c) $\frac{BC}{EF} = \frac{DM}{AL}$ (d) $\frac{AL}{DM} = \frac{AB^2}{DE^2}$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (b) $\frac{AB}{DE} = \frac{AL}{DM}$

We know that corresponding sides and corresponding medians of similar triangles are in the same ratio:

Formula / Theorem Used: $\triangle ABC \sim \triangle DEF \implies \frac{AB}{DE} = \frac{BC}{EF} = \frac{2BL}{2EM} = \frac{BL}{EM}$ and $\angle B = \angle E \implies \triangle ABL \sim \triangle DEM$ (SAS)

Therefore, corresponding sides of $\triangle ABL$ and $\triangle DEM$ satisfy $\boxed{\frac{AB}{DE} = \frac{AL}{DM}}$.

Q2. In $\triangle ABC$, $DE \parallel BC$ intersects AB at D and AC at E. If $AD = 1.5$ cm, $BD = 4.5$ cm, and $BC = 8$ cm, then DE is: [1]

(a) 2 cm (b) 2.5 cm (c) 3 cm (d) 1.8 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (a) 2 cm

Given: $AD = 1.5$ cm, $BD = 4.5$ cm (so $AB = 1.5 + 4.5 = 6.0$ cm), $BC = 8$ cm, and $DE \parallel BC$.
Since $DE \parallel BC$, $\triangle ADE \sim \triangle ABC$ (AA similarity: $\angle A$ common, $\angle ADE = \angle B$):

Formula / Theorem Used: $\frac{AD}{AB} = \frac{DE}{BC}$

$$\frac{1.5}{6.0} = \frac{DE}{8} \implies \frac{1}{4} = \frac{DE}{8} \implies DE = \frac{8}{4} = \boxed{2 \text{ cm}}$$

Q3. If a line divides two sides of a triangle in the same ratio, then the line is parallel to the third side. This theorem is: [1]

(a) Thales' Theorem (b) Converse of Thales' Theorem (c) Pythagoras Theorem (d) Midpoint Theorem

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (b) Converse of Thales' Theorem

We know Theorem 6.2 (NCERT Class X):

Formula / Theorem Used: Converse of Basic Proportionality Theorem (Converse of Thales' Theorem): If a line divides any two sides of a triangle in the same ratio, the line is parallel to the third side.

Hence, the given statement is the **Converse of Thales' Theorem**.

Q4. Which of the following pairs of triangles are **always** similar? [1]

(a) Two right triangles (b) Two isosceles right triangles (c) Two obtuse triangles (d) Two scalene triangles

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (b) Two isosceles right triangles

We know the AAA Similarity Criterion:

Formula / Theorem Used: Every isosceles right-angled triangle always has interior angles $90°$, $45°$, and $45°$.

Because all isosceles right triangles share identical angle triples $(90°, 45°, 45°)$, any two of them are always equiangular and hence similar by AAA.

Q5. In $\triangle ABC$, $DE \parallel BC$ such that $AD = 2.4$ cm, $AE = 3.2$ cm, and $EC = 4.8$ cm. The length of side AB is: [1]

(a) 6.0 cm (b) 3.6 cm (c) 7.2 cm (d) 5.4 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (a) 6.0 cm

We know the Basic Proportionality Theorem ($DE \parallel BC \implies \dfrac{AD}{BD} = \dfrac{AE}{EC}$):

Formula / Theorem Used: $\dfrac{2.4}{BD} = \dfrac{3.2}{4.8}$

Since $\dfrac{3.2}{4.8} = \dfrac{32}{48} = \dfrac{2}{3}$:

$$\frac{2.4}{BD} = \frac{2}{3} \implies BD = \frac{2.4 \times 3}{2} = 3.6 \text{ cm} \implies AB = AD + BD = 2.4 + 3.6 = \boxed{\mathbf{6.0}\text{ cm}}$$

Q6. **Assertion (A):** In right triangle PQR with $\angle Q = 90^\circ$ and $QS \perp PR$, $QS^2 = PS \cdot SR$.
Reason (R): The altitude to the hypotenuse divides the triangle into two triangles that are similar to each other. [1]

(a) Both (A) and (R) are true and (R) is the correct explanation of (A).

(b) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

(c) Assertion (A) is true but Reason (R) is false.

(d) Assertion (A) is false but Reason (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(a) Both (A) and (R) are true and (R) is the correct explanation of (A).**

We know Theorem 6.7 (Altitude on Hypotenuse):

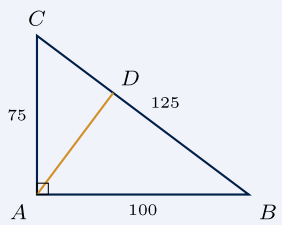
Formula / Theorem Used: $\triangle PSQ \sim \triangle QSR \implies \dfrac{PS}{QS} = \dfrac{QS}{SR} \implies QS^2 = PS \cdot SR$

Since $QS \perp PR$ in right $\triangle PQR$, the two smaller right triangles $\triangle PSQ$ and $\triangle QSR$ are similar to each other, which directly yields $QS^2 = PS \cdot SR$. Both (A) and (R) are true and (R) explains (A).

SECTION B: Very Short Answer Questions — MATHS MANTHAN | (2 × 2 = 4 Marks)

Q7. In the given figure, $\angle CAB = 90^\circ$ and $AD \perp BC$. If $AC = 75$ cm, $AB = 100$ cm, and $BC = 125$ cm, find the length of altitude AD. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]



Given: $\triangle CAB$ has $\angle CAB = 90^\circ$, $AD \perp BC$, $AC = 75$ cm, $AB = 100$ cm, $BC = 125$ cm.
We know the AA Similarity Criterion (and triangle area equivalence):

Formula / Theorem Used: $\triangle ADC \sim \triangle BAC \implies \dfrac{AD}{AB} = \dfrac{AC}{BC}$

Step 1: Establishing $\triangle ADC \sim \triangle BAC$ [1 Mark]
In $\triangle ADC$ and $\triangle BAC$: $\angle C = \angle C$ (common) and $\angle ADC = \angle BAC = 90^\circ$. By AA similarity, $\triangle ADC \sim \triangle BAC$.

Step 2: Calculating Altitude AD [1 Mark]

$$\frac{AD}{AB} = \frac{AC}{BC} \implies \frac{AD}{100} = \frac{75}{125} = \frac{3}{5} \implies AD = \frac{3}{5} \times 100 = \boxed{60 \text{ cm}}$$

(*Aliter:* $\frac{1}{2} \cdot BC \cdot AD = \frac{1}{2} \cdot AB \cdot AC \implies AD = \frac{100 \times 75}{125} = 60$ cm.)

Q8. In $\triangle ABC$ and $\triangle DEF$, $\dfrac{AB}{DE} = \dfrac{BC}{EF} = \dfrac{CA}{FD} = \dfrac{3}{4}$. If the perimeter of $\triangle ABC$ is 27 cm, determine the perimeter of $\triangle DEF$. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Given: $\dfrac{AB}{DE} = \dfrac{BC}{EF} = \dfrac{CA}{FD} = \dfrac{3}{4}$ and $\text{Perimeter}(\triangle ABC) = 27$ cm.
We know the SSS Similarity Criterion and Perimeter Property:

Formula / Theorem Used: $\dfrac{\text{Perimeter}(\triangle ABC)}{\text{Perimeter}(\triangle DEF)} = \dfrac{AB+BC+CA}{DE+EF+FD} = \dfrac{AB}{DE}$

Step 1: Applying Ratio Property [1 Mark]
Since $\triangle ABC \sim \triangle DEF$ (by SSS), each side satisfies $AB = \frac{3}{4}DE$, $BC = \frac{3}{4}EF$, $CA = \frac{3}{4}FD$, so:

$$\frac{\text{Perimeter}(\triangle ABC)}{\text{Perimeter}(\triangle DEF)} = \frac{3}{4}$$

Step 2: Finding Perimeter of $\triangle DEF$ [1 Mark]

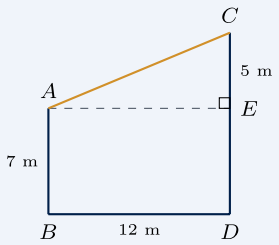
$$\frac{27}{\text{Perimeter}(\triangle DEF)} = \frac{3}{4} \implies \text{Perimeter}(\triangle DEF) = \frac{27 \times 4}{3} = 9 \times 4 = \boxed{\mathbf{36}\text{ cm}}$$

SECTION C: Short Answer Questions — MATHS MANTHAN | (2 × 3 = 6 Marks)

Q9. Two trees of heights 7 m and 12 m stand on level ground. If their bases are 12 m apart, find the straight-line distance between their tops. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: Vertical trees $AB = 7$ m and $CD = 12$ m with base distance $BD = 12$ m.
Construction: Draw $AE \perp CD$ from top A to tree CD, forming rectangle $ABDE$ and right-angled $\triangle AEC$.
We know Pythagoras Theorem for right-angled triangles:



Formula / Theorem Used: $AC^2 = AE^2 + CE^2$

Step 1: Determining Sides of Right $\triangle AEC$ [1.5 Marks]
Since $ABDE$ is a rectangle: $AE = BD = 12$ m and $ED = AB = 7$ m.
Therefore, vertical difference: $CE = CD - ED = 12 - 7 = 5$ m.

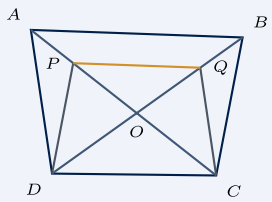
Step 2: Applying Pythagoras Theorem [1.5 Marks]
In right-angled $\triangle AEC$ ($\angle AEC = 90°$):

$$AC^2 = 12^2 + 5^2 = 144 + 25 = 169$$

$$AC = \sqrt{169} = 13 \text{ m} \implies \boxed{\text{Distance between tops } = \mathbf{13}\text{ m}}$$

Q10. In quadrilateral $ABCD$, a line through D parallel to BC meets AC at P, and a line through C parallel to AD meets BD at Q. Prove that: $PQ \parallel AB$. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]



Given: Quadrilateral $ABCD$ with diagonals AC and BD intersecting at O; $DP \parallel BC$ (meets AC at P) and $CQ \parallel AD$ (meets BD at Q).
To Prove: $PQ \parallel AB$.
We know the AA Similarity Criterion and Converse of BPT:

Formula / Theorem Used: If $\dfrac{OP}{OA} = \dfrac{OQ}{OB}$ in $\triangle OAB$, then $PQ \parallel AB$.

Step 1: Using $DP \parallel BC$ ($\triangle ODP \sim \triangle OBC$) [1 Mark]
In $\triangle ODP$ and $\triangle OBC$: $\angle DOP = \angle BOC$ (vertically opposite) and $\angle ODP = \angle OBC$ (alternate interior angles, $DP \parallel BC$). By AA similarity, $\triangle ODP \sim \triangle OBC$:

$$\frac{OP}{OC} = \frac{OD}{OB} \implies OP \cdot OB = OC \cdot OD \qquad \text{— (i)}$$

Step 2: Using $CQ \parallel AD$ ($\triangle OAD \sim \triangle OCQ$) [1 Mark]
In $\triangle OAD$ and $\triangle OCQ$: $\angle AOD = \angle COQ$ (vertically opposite) and $\angle OAD = \angle OCQ$ (alternate interior angles, $CQ \parallel AD$). By AA similarity, $\triangle OAD \sim \triangle OCQ$:

$$\frac{OA}{OC} = \frac{OD}{OQ} \implies OA \cdot OQ = OC \cdot OD \qquad \text{— (ii)}$$

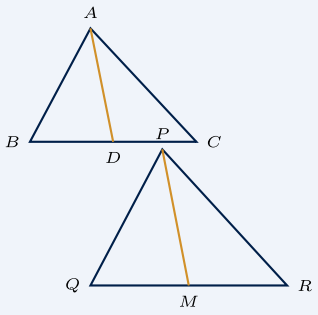
Step 3: Applying Converse of BPT in $\triangle OAB$ [1 Mark]
Equating (i) and (ii): $OP \cdot OB = OA \cdot OQ \implies \dfrac{OP}{OA} = \dfrac{OQ}{OB}$.
By the **Converse of BPT** in $\triangle OAB$:

$$\boxed{\therefore PQ \parallel AB \quad \text{[Hence Proved]}}$$

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Q11. Prove that if two sides and a median bisecting one of these sides of a triangle are respectively proportional to the two sides and the corresponding median of another triangle, then the two triangles are similar ($\triangle ABC \sim \triangle PQR$ given $\frac{AB}{PQ} = \frac{BC}{QR} = \frac{AD}{PM}$). **[5]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]



Given: In $\triangle ABC$ and $\triangle PQR$, AD and PM are medians to BC and QR respectively ($BD = \frac{1}{2}BC$, $QM = \frac{1}{2}QR$) and $\frac{AB}{PQ} = \frac{BC}{QR} = \frac{AD}{PM}$.

To Prove: $\triangle ABC \sim \triangle PQR$.

We know the SSS and SAS similarity criteria:

Formula / Theorem Used: Halving proportional sides preserves proportionality, enabling SSS similarity of half-triangles to recover the included angle.

Step 1: Establishing $\triangle ABD \sim \triangle PQM$ [2.5 Marks]

Since D is midpoint of BC and M is midpoint of QR:

$$\frac{BC}{QR} = \frac{2BD}{2QM} = \frac{BD}{QM}$$

Substituting into the given proportion:

$$\frac{AB}{PQ} = \frac{BD}{QM} = \frac{AD}{PM}$$

In $\triangle ABD$ and $\triangle PQM$, all three pairs of corresponding sides are proportional. By **SSS similarity criterion**, $\triangle ABD \sim \triangle PQM$.

Therefore, corresponding angles are equal: $\angle B = \angle Q$ — (i)

Step 2: Proving $\triangle ABC \sim \triangle PQR$ by SAS [2.5 Marks]

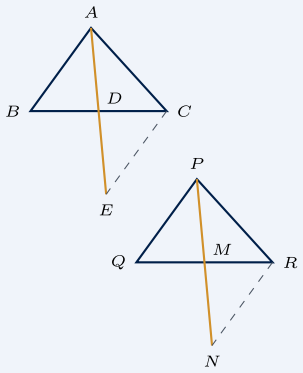
In $\triangle ABC$ and $\triangle PQR$: (1) $\frac{AB}{PQ} = \frac{BC}{QR}$ (Given), and (2) $\angle B = \angle Q$ (From (i)).

Since two pairs of sides are proportional and the included angles are equal, by the **SAS similarity criterion**:

$$\therefore \triangle ABC \sim \triangle PQR \quad \text{[Hence Proved]}$$

Q12. Sides AB and AC and median AD of a triangle ABC are respectively proportional to sides PQ and PR and median PM of another triangle PQR. Prove that: $\triangle ABC \sim \triangle PQR$. **[5]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]



Given: In $\triangle ABC$ and $\triangle PQR$, AD and PM are medians to BC and QR respectively, and $\frac{AB}{PQ} = \frac{AC}{PR} = \frac{AD}{PM}$. **To Prove:** $\triangle ABC \sim \triangle PQR$.

Construction: Produce AD to E such that $AD = DE$, join CE; produce PM to N such that $PM = MN$, join RN.

We know the SAS congruence, SSS similarity, and SAS similarity criteria:

Formula / Theorem Used: Doubling medians creates triangles ($\triangle ACE \sim \triangle PRN$) sharing halves of the apex angle.

Step 1: Congruent Triangles via Construction [1.5 Marks]

In $\triangle ABD$ and $\triangle ECD$: $BD = CD$ (median), $\angle ADB = \angle EDC$ (vert. opp.), $AD = ED$ (const.) $\implies \triangle ABD \cong \triangle ECD$ (SAS) $\implies CE = AB$.

Similarly, $\triangle PQM \cong \triangle NRM$ (SAS) $\implies RN = PQ$.

Step 2: Establishing $\triangle ACE \sim \triangle PRN$ [2 Marks]

Using $CE = AB$, $RN = PQ$, $AE = 2AD$, $PN = 2PM$:

$$\frac{AB}{PQ} = \frac{AC}{PR} = \frac{AD}{PM} \implies \frac{CE}{RN} = \frac{AC}{PR} = \frac{2AD}{2PM} = \frac{AE}{PN}$$

By **SSS similarity**, $\triangle ACE \sim \triangle PRN \implies \angle CAE = \angle RPN$ (i.e., $\angle CAD = \angle RPM$).

By symmetry (joining BE, QN): $\angle BAD = \angle QPM$.

Step 3: Proving $\triangle ABC \sim \triangle PQR$ by SAS [1.5 Marks]

Adding the two angle equalities: $\angle BAD + \angle CAD = \angle QPM + \angle RPM \implies \angle A = \angle P$.

In $\triangle ABC$ and $\triangle PQR$: $\frac{AB}{PQ} = \frac{AC}{PR}$ (given) and $\angle A = \angle P$ (proved). By **SAS similarity**:

$$\therefore \triangle ABC \sim \triangle PQR \quad \text{[Hence Proved]}$$

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Q13. Case Study 09: School Sports Ground Flagpole Calibration & Scale Factor

During a mathematics lab activity, Class 10 students are tasked with calibrating a scale model for the school sports complex. The school flagpole is 15 m high. At 10:00 AM, its shadow measures 6 m. Nearby, a student of height 1.5 m casts a shadow x m. [4]

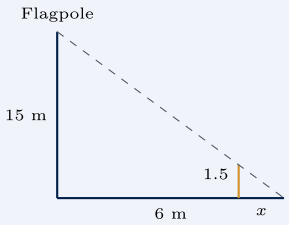
(i) **[1 Mark]** Find the exact length of the student's shadow x.

(ii) **[1 Mark]** If a basketball hoop pole casts a shadow of 1.2 m at that same instant, find the height of the hoop pole.

(iii) **[2 Marks]** If a 3D scale model is built with scale factor 1 : 50, find the height of the flagpole in the model and state the ratio of perimeters of any similar component.

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]

Given Data: Flagpole height $AB = 15$ m, shadow $BC = 6$ m; student height $DE = 1.5$ m, shadow $EC = x$ m.
At the same instant (10:00 AM), sun rays fall at equal angles ($\triangle ABC \sim \triangle DEC$ by AA similarity):



Formula / Theorem Used: $\dfrac{\text{Object Height}}{\text{Shadow Length}} = \dfrac{15}{6} = 2.5$

Sub-part (i): Exact Length of Student's Shadow x [1 Mark]
Since shadow triangles are similar:

$$\frac{15}{6} = \frac{1.5}{x} \implies x = \frac{1.5 \times 6}{15} = \frac{9.0}{15} = 0.6 \text{ m} \implies \boxed{x = \mathbf{0.6} \text{ m}}$$

Sub-part (ii): Height of Basketball Hoop Pole (h) [1 Mark]
With shadow 1.2 m at the same 10:00 AM elevation ratio:

$$\frac{h}{1.2} = \frac{15}{6} = 2.5 \implies h = 2.5 \times 1.2 = 3.0 \text{ m} \implies \boxed{\text{Height of hoop pole } = \mathbf{3.0} \text{ m}}$$

Sub-part (iii): Model Flagpole Height & Perimeter Ratio (Scale Factor 1 : 50) [2 Marks]
1. Scale factor $k = \dfrac{1}{50}$. Model flagpole height $= \dfrac{1}{50} \times 15$ m $= 0.3$ m $= 30$ cm.
2. For any two similar figures, the ratio of their perimeters equals the linear scale factor $= 1 : 50$.

$$\boxed{\text{Model Flagpole Height } = \mathbf{0.3} \text{ m } (\mathbf{30} \text{ cm}), \qquad \text{Perimeter Ratio } = \mathbf{1 : 50}}$$

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